UNIT II

Micro & Macro Mechanics of a Lamina

Rule of mixtures | four elastic moduli | Hooke's law | compliance & stiffness | angle lamina

I. Micro Mechanical Analysis of a Lamina

Micromechanics studies the behaviour of a lamina by considering the **individual properties and interaction of fibre and matrix**. Its aim is to predict the lamina (ply) properties E_1, E_2, ν_{12}, G_{12} from fibre and matrix properties and the volume fractions.

Term	Meaning
Lamina (ply)	Single flat layer of unidirectional / woven fibres in a matrix; basic building block, thickness 0.1-0.25 mm
Laminate	Stack of laminae bonded together at different orientations (Unit III)
Unidirectional (UD)	All fibres parallel; highest E_1, strongly orthotropic
Woven / fabric	Fibres in warp & weft; nearly equal strength in 2 directions (balanced weave)
Principal material axes	1 = along fibres, 2 = in-plane transverse, 3 = thickness

Approaches to find lamina properties

> **Mechanics of materials** (rule of mixtures) - simple assumptions on strain / stress; used in this unit.

> **Elasticity** (bounds, cylindrical assembly, Hashin) and **semi-empirical** (Halpin-Tsai) models.

> **Experiments** (tension, compression, V-notch shear tests) remain final authority.

I.1 Basic assumptions

- Lamina is macroscopically homogeneous, linearly elastic and **orthotropic**.
- Fibres are uniform, parallel and continuous; perfect fibre-matrix bond (no slip).
- Matrix is void-free, isotropic and linearly elastic; fibres also linearly elastic.
- Initially stress-free lamina; loads applied in principal material directions.

I.2 Volume fraction, weight fraction, density

Let v = volume, w = weight, ρ = density; subscripts c, f, m = composite, fibre, matrix.

$$V_f = v_f / v_c \,,\; V_m = v_m / v_c \,,\; V_f + V_m + V_v = 1$$
$$w_f = w_f / w_c \,,\; w_f + w_m = 1$$
$$\rho_c = \rho_f V_f + \rho_m V_m \,,\; 1/\rho_c = w_f/\rho_f + w_m/\rho_m$$
$$w_f = \rho_f V_f / \rho_c \,,\; V_f = (w_f/\rho_f) / (w_f/\rho_f + w_m/\rho_m)$$

Voids: $V_v = (\rho_t - \rho_e) / \rho_t$, where ρ_t = theoretical and ρ_e = experimental density. Even 1 % voids can cut inter-laminar shear strength by about 7 %.

II. Evaluation of the Four Elastic Moduli (Rule of Mixtures)

A lamina has four independent in-plane constants: E_1 (longitudinal modulus), E_2 (transverse modulus), ν_{12} (major Poisson's ratio) and G_{12} (in-plane shear modulus). They are found by the **strength-of-materials approach**.

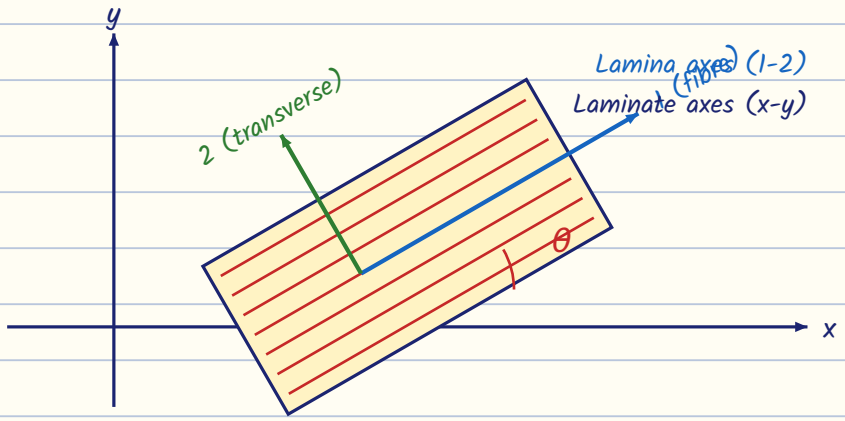
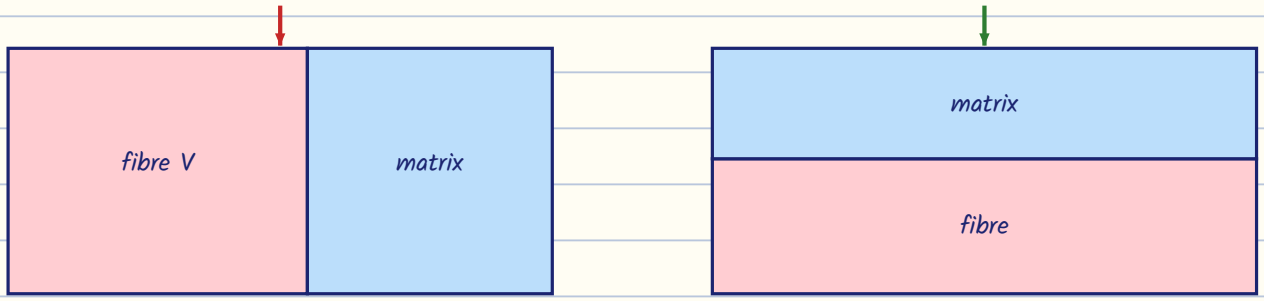


Fig. 1 Lamina with principal material axes 1-2



Voigt: equal strain → E1 **Reuss: equal stress → E2**

Fig. 2 Mechanics models: parallel (E1, 12) and series (E2, G12)

2.1 Longitudinal modulus E1 (equal strain model)

Load P is applied along the fibres. Fibre and matrix **strain equally**: $\varepsilon_c = \varepsilon_f = \varepsilon_m$. The load is shared: $P_c = P_f + P_m$.

$$\sigma_c A_c = \sigma_f A_f + \sigma_m A_m \Rightarrow E_c \varepsilon A_c = E_f \varepsilon A_f + E_m \varepsilon A_m$$
$$E_1 = E_f V_f + E_m V_m$$

E_1 is the **upper bound** (Voigt) and is very accurate (within 1-2 % of tests). Fibres carry the fraction $P_f/P_c = E_f V_f / E_1$.

Derivation of E1 - step by step

- Take a lamina element of length l, area $A_c = A_f + A_m$; volume fractions $V_f = A_f/A_c$, $V_m = A_m/A_c$ (length equal).
- Equilibrium: $P_c = P_f + P_m \rightarrow \sigma_1 A_c = \sigma_f A_f + \sigma_m A_m \rightarrow \sigma_1 = \sigma_f V_f + \sigma_m V_m$.
- Compatibility: the fibre, matrix and composite elongate equally, $\varepsilon_1 = \varepsilon_f = \varepsilon_m$.
- Constitutive: $\sigma_f = E_f \varepsilon_1$, $\sigma_m = E_m \varepsilon_1$; substitute $\rightarrow \sigma_1 = (E_f V_f + E_m V_m)\, \varepsilon_1 \rightarrow E_1 = \sigma_1/\varepsilon_1$.

2.2 Transverse modulus E2 (equal stress model)

Load is applied **perpendicular** to fibres. Both constituents carry the **same stress** σ, but strain differently; the total deformation is the sum: $\delta_c = \delta_f + \delta_m$, i.e. $\varepsilon_2 t_c = \varepsilon_f t_f + \varepsilon_m t_m$.

$$\sigma/E_2 = V_f\, \sigma/E_f + V_m\, \sigma/E_m$$
$$1/E_2 = V_f/E_f + V_m/E_m \text{ or } E_2 = E_f E_m / (E_f V_m + E_m V_f)$$

This is the **lower bound** (Reuss). It under-predicts test values; the **Halpin-Tsai** semi-empirical equation ($E_2/E_m = (1 + \xi\eta V_f)/(1 - \eta V_f)$, $\eta = (E_f/E_m - 1)/(E_f/E_m + \xi)$) gives a better fit.

2.3 Major Poisson's ratio 12

Poisson's ratio = - (transverse strain)/(longitudinal strain) when loaded along fibres. Transverse deformation of the lamina is the sum of those of fibre and matrix (equal longitudinal strain ε_1):

$$\nu_{12} = \nu_f V_f + \nu_m V_m \text{ and } \nu_{21} = \nu_{12}\, E_2/E_1$$

2.4 In-plane shear modulus G12

Shear stress τ acts on fibre and matrix equally (series model) and shear deformations add: $\gamma_c t_c = \gamma_f t_f + \gamma_m t_m$.

$$1/G_{12} = V_f/G_f + V_m/G_m \Rightarrow G_{12} = G_f G_m / (G_f V_m + G_m V_f)$$
with $G = E / 2(1 + \nu)$ for each isotropic constituent

Constant	Rule of mixtures	Remark
E_1	$E_f V_f + E_m V_m$	Voigt; fibre dominated; accurate
E_2	$E_f E_m / (E_f V_m + E_m V_f)$	Reuss; matrix dominated; low

Constant	Rule of mixtures	Remark
ν_{12}	$\nu_f V_f + \nu_m V_m$	$\nu_{21} = \nu_{12} E_2 / E_1$
G_{12}	$G_f G_m / (G_f V_m + G_m V_f)$	Matrix dominated; under-predicts

2.5 Effect of fibre volume fraction

V_f	0.2	0.3	0.4	0.5	0.6	0.7
E_1 (GPa)	49.4	72.1	94.8	116.8	139.4	162.1
E_2 (GPa)	4.3	4.8	5.7	6.9	8.6	11.4

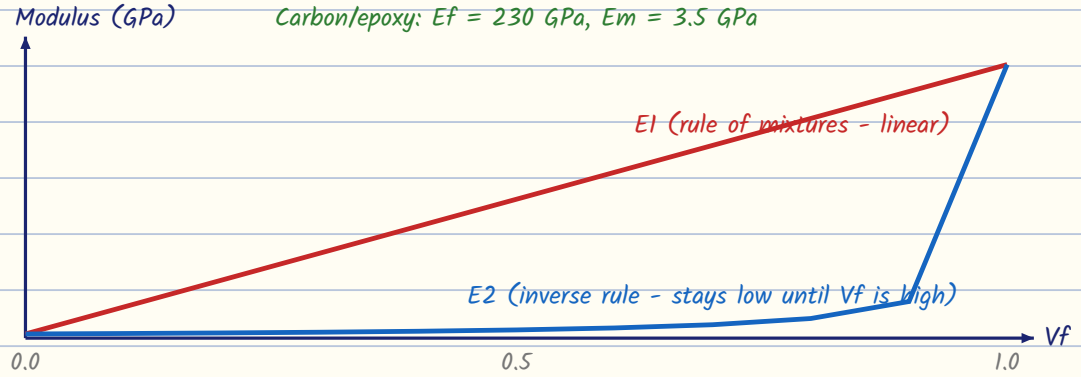


Fig. 3 Variation of E1 and E2 with Vf (carbon/epoxy)

- E_1 varies **linearly** with V_f; to double E_1 we need to double V_f.
- E_2 is controlled by the **matrix**: it rises only slowly until V_f is large - so transverse stiffness cannot be bought with more fibre.
- Practical limit $V_f \approx 0.65$–0.70 (hexagonal packing limit 0.907, square 0.785).
- Halpin-Tsai with $\xi = 2$ for E_2 gives a more realistic (higher) value than the inverse rule.

III. Numerical Problems on Micromechanics

Problem 1

Glass/epoxy lamina: $E_f = 72$ GPa, $E_m = 3.5$ GPa, $G_f = 30$ GPa, $G_m = 1.3$ GPa, $\nu_f = 0.22$, $\nu_m = 0.35$, $V_f = 0.6$. Find E_1, E_2, ν_{12}, G_{12}.

$V_m = 1 - 0.6 = 0.4$

$E_1 = 72(0.6) + 3.5(0.4) = 43.2 + 1.4 = \mathbf{44.6\ GPa}$

$E_2 = (72 \times 3.5) / (72 \times 0.4 + 3.5 \times 0.6) = 252 / 30.9 = \mathbf{8.16\ GPa}$

$\nu_{12} = 0.22(0.6) + 0.35(0.4) = 0.132 + 0.14 = \mathbf{0.272}$

$G_{12} = (30 \times 1.3) / (30 \times 0.4 + 1.3 \times 0.6) = 39 / 12.78 = \mathbf{3.05\ GPa}$

Problem 2

A glass/epoxy composite has weight fraction of glass $W_f = 0.7$, $\rho_f = 2.5$ g/cc, $\rho_m = 1.2$ g/cc. Find V_f and ρ_c.

$1/\rho_c = 0.7/2.5 + 0.3/1.2 = 0.28 + 0.25 = 0.53 \Rightarrow \rho_c =$ **1.887 g/cc**

$V_f = \rho_c W_f/\rho_f = 1.887 \times 0.7 / 2.5 =$ **0.528** (52.8 %)

Problem 3

Carbon/epoxy: $E_f = 230$ GPa, $E_m = 3.5$ GPa, $V_f = 0.5$. Find E_1, E_2 and the load share of fibres along 1.

$E_1 = 230(0.5) + 3.5(0.5) =$ **116.75 GPa**; $E_2 = 230 \times 3.5 / (230 \times 0.5 + 3.5 \times 0.5) = 805 / 116.75$ = **6.90 GPa**

Fibre load share = $E_f V_f/E_1 = 115/116.75 =$ **98.5 %** ($E_1/E_2 \approx 17$: strongly orthotropic).

Problem 7

Carbon/epoxy ($E_f = 230$, $E_m = 3.5$ GPa). What V_f is needed for $E_1 = 100$ GPa? What strain does $\sigma_1 = 100$ MPa produce?

$100 = 230\ V_f + 3.5(1 - V_f) \Rightarrow V_f = 96.5 / 226.5 =$ **0.426**

$\varepsilon_1 = \sigma_1/E_1 = 100$ MPa / 100 GPa = **1000** $\mu\varepsilon$ (0.1 %) ; fibre stress = $E_f\varepsilon_1 =$ **230 MPa**, matrix stress = **3.5 MPa**.

Problem 8

A lamina has $E_1 = 140$ GPa, $E_2 = 10$ GPa, $\nu_{12} = 0.3$. Find ν_{21} and check the stability condition.

$\nu_{21} = \nu_{12}E_2/E_1 = 0.3 \times 10/140 =$ **0.0214**. Limit: $|\nu_{12}| < \sqrt{(E_1/E_2)} = \sqrt{14} = 3.74 \rightarrow 0.3 < 3.74$ ✓ (valid).

IV. Macro Mechanics of a Lamina

Macromechanics treats the lamina as a **homogeneous, anisotropic** continuum and uses average (apparent) properties E_1, E_2, ν_{12}, G_{12} to find stress-strain relations; the fibre and matrix are not separately visible.

4.1 Generalised Hooke's law

For a 3-D linearly elastic body the stress components σ_i ($\sigma_1 \ldots \sigma_3$ normal, τ_{23}, τ_{31}, τ_{12} shear) and strains ε_j are related by (contracted notation):

$$\{\sigma_i\} = [C_{ij}]\{\varepsilon_j\} \text{ (stiffness) } \{\varepsilon_i\} = [S_{ij}]\{\sigma_j\} \text{ (compliance)}$$
$$[S] = [C]^{-1} \; ; \; i, j = 1 \dots 6$$

The 81 constants of the 4th-order tensor reduce to **36** by symmetry of stress and strain, and to **21** (symmetric matrix $C_{ij} = C_{ji}$) by the existence of a **strain-energy function**.

Engineering constants are defined from simple tests: $E_i = \sigma_i/\varepsilon_i$ (tension), $\nu_{ij} = -\varepsilon_j/\varepsilon_i$ (load in i), $G_{ij} = \tau_{ij}/\gamma_{ij}$ (shear). The compliance matrix of an **orthotropic** material (9 constants, 12 non-zero terms, 3 pairs equal) has the form:

$$\begin{bmatrix} 1/E1 & -\nu21/E2 & -\nu31/E3 & 0 & 0 & 0 \\ -\nu12/E1 & 1/E2 & -\nu32/E3 & 0 & 0 & 0 \\ -\nu13/E1 & -\nu23/E2 & 1/E3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G23 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G31 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G12 \end{bmatrix}$$

Orthotropic compliance [S]: normal-shear coupling terms are zero

4.2 Number of elastic constants for different materials

Material	Symmetry planes	Independent constants
Fully anisotropic (triclinic)	None	21
Monoclinic	One plane of symmetry	13
Orthotropic (lamina, wood)	Three mutually ⊥ planes	9 ($E_1, E_2, E_3, \nu_{12}, \nu_{13}, \nu_{23}, G_{12}, G_{13}, G_{23}$)
Transversely isotropic (UD fibres)	Isotropic in plane 2-3	5 ($E_1, E_2, \nu_{12}, \nu_{23}, G_{12}$)
Isotropic (metals)	Infinite planes	2 (E, ν); $G = E/2(1+\nu)$

Remember: In the orthotropic case **no coupling** exists between normal stresses and shear strains ($S_{16} = S_{26} = 0$) when loaded along the material axes.

4.3 Plane stress: two-dimensional relations

A thin lamina has $\sigma_3 = \tau_{23} = \tau_{31} = 0$ (**plane stress**), so only σ_1, σ_2, τ_{12} remain.

$$\{\varepsilon_1 \; \varepsilon_2 \; \gamma_{12}\}^T = [S]\{\sigma_1 \; \sigma_2 \; \tau_{12}\}^T$$
$$S_{11} = 1/E_1 \, , \; S_{22} = 1/E_2 \, , \; S_{12} = -\nu_{12}/E_1 = -\nu_{21}/E_2 \, , \; S_{66} = 1/G_{12}$$

Inverting gives the **reduced stiffness matrix** $[Q] = [S]^{-1}$:

$$Q_{11} = E_1/(1 - \nu_{12}\nu_{21}) \, , \; Q_{22} = E_2/(1 - \nu_{12}\nu_{21})$$
$$Q_{12} = \nu_{12}E_2/(1 - \nu_{12}\nu_{21}) \, , \; Q_{66} = G_{12}$$

Maxwell-Betti reciprocity: $\nu_{12}/E_1 = \nu_{21}/E_2$. Positive definiteness requires $E, G > 0$ and $|\nu_{12}| < \sqrt{(E_1/E_2)}$.

V. Hooke's Law for a Two-Dimensional Angle Lamina

When fibres make an angle θ with the loading axis x (**off-axis / angle lamina**), stresses and strains in x-y are related by the **transformed** matrices. Shear-extension coupling appears: a normal stress produces shear strain.

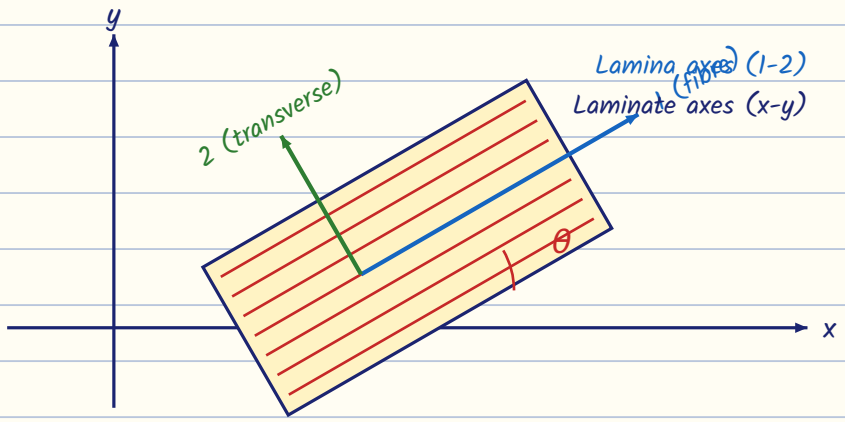


Fig. 4 Angle lamina: material axes (1,2) rotated by from x-y

5.1 Transformation of stresses and strains

$$\{\sigma_1\ \sigma_2\ \tau_{12}\}^T = [T]\ \{\sigma_x\ \sigma_y\ \tau_{xy}\}^T$$

$$[T] = [\ m^2\ n^2\ 2mn\ ;\ n^2\ m^2\ -2mn\ ;\ -mn\ mn\ m^2 - n^2\ ]$$

$$m = \cos\theta\ ,\ n = \sin\theta$$

Strains transform the same way with the engineering shear strain γ: use the Reuter matrix $[R] = \text{diag}(1, 1, 2)$, giving ε-transformation $[R][T][R]^{-1} = [T]^{-T}$.

Derivation idea (wedge element)

- Cut a small triangular element from the lamina with its inclined face normal to axis 1 (angle θ from x).
- Write force equilibrium along 1 and 2 using areas A·cosθ, A·sinθ → gives the [T] matrix ($\sigma_1 = \sigma_x m^2 + \sigma_y n^2 + 2\tau_{xy} mn$ ...).
- Strains transform by the same geometry but with tensor shear γ/2, hence the Reuter matrix [R].
- Substituting $\{\sigma\} = [Q]\{\varepsilon\}$: $\{\sigma_{xy}\} = [T]^{-1}[Q][T]^{-T}\{\varepsilon_{xy}\} = [\bar{Q}]\{\varepsilon_{xy}\}$.

5.2 Transformed reduced stiffness [Q] and compliance [S]

$$\{\sigma_x\ \sigma_y\ \tau_{xy}\}^T = [\bar{Q}]\ \{\varepsilon_x\ \varepsilon_y\ \gamma_{xy}\}^T,\quad [\bar{Q}] = [T]^{-1}[Q][R][T][R]^{-1}$$
$$\{\varepsilon_x\ \varepsilon_y\ \gamma_{xy}\}^T = [\bar{S}]\ \{\sigma_x\ \sigma_y\ \tau_{xy}\}^T,\quad [\bar{S}] = [\bar{Q}]^{-1}$$

Expanded forms ($m = \cos\theta$, $n = \sin\theta$):

$$\bar{Q}_{11} = Q_{11}m^4 + 2(Q_{12} + 2Q_{66})m^2n^2 + Q_{22}n^4$$
$$\bar{Q}_{12} = (Q_{11} + Q_{22} - 4Q_{66})m^2n^2 + Q_{12}(m^4 + n^4)$$
$$\bar{Q}_{22} = Q_{11}n^4 + 2(Q_{12} + 2Q_{66})m^2n^2 + Q_{22}m^4$$
$$\bar{Q}_{16} = (Q_{11} - Q_{12} - 2Q_{66})m^3n + (Q_{12} - Q_{22} + 2Q_{66})mn^3$$
$$\bar{Q}_{26} = (Q_{11} - Q_{12} - 2Q_{66})mn^3 + (Q_{12} - Q_{22} + 2Q_{66})m^3n$$
$$\bar{Q}_{66} = (Q_{11} + Q_{22} - 2Q_{12} - 2Q_{66})m^2n^2 + Q_{66}(m^4 + n^4)$$

Properties of [Q]

- Fully populated (9 non-zero terms) but only **4 independent** constants (+ θ).
- Symmetric: $\bar{Q}_{ij} = \bar{Q}_{ji}$.
- At $\theta = 0°$ or $90°$: $\bar{Q}_{16} = \bar{Q}_{26} = 0$ (specially orthotropic).
- $\bar{Q}_{16}$, $\bar{Q}_{26}$ are odd functions of θ: sign changes for $+\theta$ / $-\theta$.

5.3 Engineering constants of an angle lamina

$$1/E_x = m^4/E_1 + (1/G_{12} - 2\nu_{12}/E_1)\,m^2n^2 + n^4/E_2$$
$$1/E_y = n^4/E_1 + (1/G_{12} - 2\nu_{12}/E_1)\,m^2n^2 + m^4/E_2$$
$$1/G_{xy} = 2(2/E_1 + 2/E_2 + 4\nu_{12}/E_1 - 1/G_{12})\,m^2n^2 + (m^4 + n^4)/G_{12}$$
$$\nu_{xy} = E_x[\nu_{12}(m^4 + n^4)/E_1 - (1/E_1 + 1/E_2 - 1/G_{12})\,m^2n^2]$$

Coefficients of mutual influence: $\eta_{x,xy} = \varepsilon_{xy} / \varepsilon_x$ (shear strain per unit normal strain) and $\eta_{xy,x}$ - the sign of coupling follows θ.

VI. Numerical Problems - Angle Lamina

Problem 4

Glass/epoxy: $E_1 = 38.6$, $E_2 = 8.27$, $G_{12} = 4.14$ GPa, $\nu_{12} = 0.26$. Find [S], [Q] and [$\bar{Q}$] at $\theta = 45°$.

$\nu_{21} = 0.26 \times 8.27/38.6 = 0.0557$; $1 - \nu_{12}\nu_{21} = 0.9855$

[S] (1/GPa): $S_{11} = 0.02591$, $S_{12} = -0.00674$, $S_{22} = 0.12092$, $S_{66} = 0.24155$

[Q] (GPa): $Q_{11} = 39.17$, $Q_{12} = 2.18$, $Q_{22} = 8.39$, $Q_{66} = 4.14$

At 45° ($m^2 = n^2 = 0.5$): $\bar{Q}_{11} = \bar{Q}_{22} = 17.12$, $\bar{Q}_{12} = 8.84$, $\bar{Q}_{66} = 10.80$, $\bar{Q}_{16} = \bar{Q}_{26} = 7.69$ GPa.

θ	$\bar{Q}_{11}$	$\bar{Q}_{12}$	$\bar{Q}_{22}$	$\bar{Q}_{16}$	$\bar{Q}_{26}$	$\bar{Q}_{66}$
30°	26.48	7.18	11.09	9.55	3.78	9.13
45°	17.12	8.84	17.12	7.69	7.69	10.80
60°	11.09	7.18	26.48	3.78	9.55	9.13

Problem 5

Same lamina, stresses in x-y: $\sigma_x = 2$ MPa, $\sigma_y = -3$ MPa, $\tau_{xy} = 4$ MPa, $\theta = 30°$. Find stresses and strains in material axes.

$m = \cos 30° = 0.866$, $n = 0.5$. $\sigma_1 = m^2\sigma_x + n^2\sigma_y + 2mn\,\tau_{xy} = 1.5 - 0.75 + 3.464 =$ **4.214 MPa**

$\sigma_2 = n^2\sigma_x + m^2\sigma_y - 2mn\,\tau_{xy} = 0.5 - 2.25 - 3.464 =$ **-5.214 MPa**

$\tau_{12} = -mn\,\sigma_x + mn\,\sigma_y + (m^2 - n^2)\,\tau_{xy} = -0.866 - 1.299 + 2.0 =$ **-0.165 MPa**

$\varepsilon_1 = S_{11}\sigma_1 + S_{12}\sigma_2 =$ **144.3** $\mu\varepsilon$; $\varepsilon_2 = S_{12}\sigma_1 + S_{22}\sigma_2 =$ **-658.9** $\mu\varepsilon$; $\gamma_{12} = S_{66}\tau_{12} =$ **-39.9** μ

Problem 6

For the same material find E_x, E_y, G_{xy} at $\theta = 30°$ and 45°.

$\theta = 30°$: $E_x =$ **15.41 GPa**, $E_y =$ **8.90 GPa**, $G_{xy} =$ **5.54 GPa**

$\theta = 45°$: $E_x = E_y =$ **10.67 GPa**, $G_{xy} =$ **6.24 GPa** (E_x falls from 38.6 to 10.7 GPa - fibre direction matters!)

Problem 9

Glass/epoxy lamina at $\theta = 60°$ is subjected to strains $\varepsilon_x = 0.001$, $\varepsilon_y = 0$, $\gamma_{xy} = 0$. Find the stresses σ_x, σ_y, τ_{xy}.

At 60°: $\bar{Q}_{11} = 11.09$, $\bar{Q}_{12} = 7.18$, $\bar{Q}_{16} = 3.78$ GPa (from the table above).

$\sigma_x = \bar{Q}_{11}\varepsilon_x =$ **11.09 MPa** ; $\sigma_y = \bar{Q}_{12}\varepsilon_x =$ **7.18 MPa** ; $\tau_{xy} = \bar{Q}_{16}\varepsilon_x =$ **3.78 MPa**

Observe: A pure normal strain ε_x induces **shear stress** τ_{xy} - a result of anisotropy; it does not occur in isotropic materials.

VII. Stress-Strain Relations for a Lamina of Arbitrary Orientation

- **Step 1** - find E_1, E_2, ν_{12}, G_{12}; compute [S] and [Q].
- **Step 2** - set $m = \cos\theta$, $n = \sin\theta$ (θ +ve counter-clockwise from x to fibre direction 1).
- **Step 3** - get $[\bar{Q}]$ by formula, or $[S] = [\bar{Q}]^{-1}$.
- **Step 4** - given σ_x, σ_y, $\tau_{xy} \rightarrow \varepsilon_x$, ε_y, $\gamma_{xy} = [S]\{\sigma\}$; or given strains → stresses.
- **Step 5** - transform to material axes ($\{\sigma_{1,2,12}\} = [T]\{\sigma_{xy}\}$) for failure checking (Unit III).

Special behaviour of angle lamina

- **Shear-extension coupling**: σ_x alone produces γ_{xy} ($S_{16} \neq 0$); a rectangular element distorts.
- **Normal-shear (η) interaction** is absent for $\theta = 0°$, $90°$, and for isotropic materials.
- E_x varies from E_1 ($\theta = 0°$) to E_2 ($\theta = 90°$): minimum may be below E_2 when G_{12} is small.

Specially orthotropic ($\theta = 0°, 90°$)	Generally orthotropic (angle lamina)
Axes x-y coincide with 1-2	Axes x-y rotated by θ from 1-2
$\bar{Q}_{16} = \bar{Q}_{26} = 0$: no shear coupling	$\bar{Q}_{16}, \bar{Q}_{26} \neq 0$: shear-extension coupling
4 constants: E_1, E_2, ν_{12}, G_{12}	4 constants + θ, with 6 non-zero $\bar{Q}_{ij}$
Used in micromechanics and tests	Used in laminate analysis (Unit III)

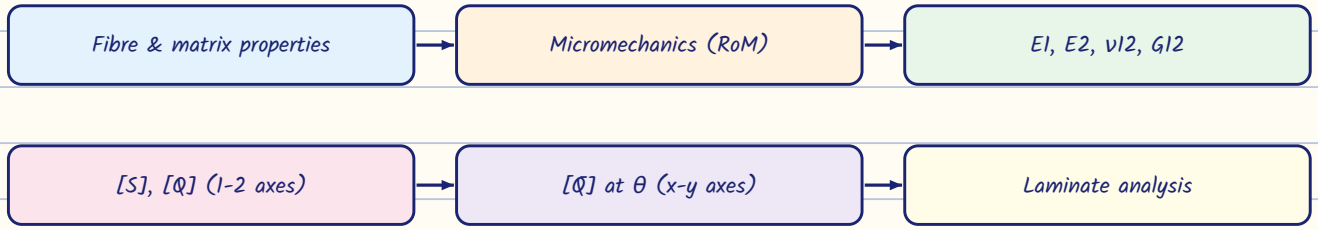


Fig. 5 Road-map from constituents to laminate

7.1 Typical unidirectional lamina properties (V_f 0.5–0.6)

Material	E_1 (GPa)	E_2 (GPa)	G_{12} (GPa)	ν_{12}
E-glass / epoxy	38.6	8.27	4.14	0.26
Kevlar-49 / epoxy	76	5.5	2.3	0.34
Carbon T300 / epoxy	181	10.3	7.17	0.28
Boron / epoxy	204	18.5	5.59	0.23

Problem 10

Carbon/epoxy ($E_1 = 181$, $E_2 = 10.3$, $G_{12} = 7.17$ GPa, $\nu_{12} = 0.28$): find E_x for fibre angles 15°, 30°, 45°, 60°.

Use $1/E_x = m^4/E_1 + (1/G_{12} - 2\nu_{12}/E_1)m^2n^2 + n^4/E_2$

θ	0°	15°	30°	45°	60°	90°
E_x (GPa)	181	72.6	28.8	16.7	12.4	10.3

Conclusion: a mere **15° misalignment** cuts the stiffness by 60 % - fibre alignment during manufacture is critical (and the reason for multi-angle laminates).

Problem II (Halpin-Tsai)

Find E_2 for carbon/epoxy (E_f = 230, E_m = 3.5 GPa, V_f = 0.5) with ξ = 2 and compare with the inverse rule of mixtures.

$\eta = (E_f/E_m - 1) / (E_f/E_m + \xi) = (65.71 - 1) / (65.71 + 2) =$ **0.956**

$E_2 = E_m(1 + \xi\eta V_f) / (1 - \eta V_f) = 3.5(1 + 0.956) / (1 - 0.478) =$ **13.1 GPa** vs 6.9 GPa (inverse RoM): Halpin-Tsai is nearly twice - closer to tests.

Viva / short questions

> Why is E_1 called the upper bound and E_2 the lower bound?
> What is the difference between a lamina and a laminate?
> How many independent constants for a UD lamina in plane stress? (4: E_1, E_2, ν_{12}, G_{12})
> Why does [Q] have 9 non-zero terms but only 4 independent constants?
> Why is $\nu_{21} \neq \nu_{12}$ for an orthotropic lamina?

VIII. Quick Revision - Unit II

Formulae to remember

> $E_1 = E_fV_f + E_mV_m$; $1/E_2 = V_f/E_f + V_m/E_m$; $\nu_{12} = \nu_fV_f + \nu_mV_m$; $1/G_{12} = V_f/G_f + V_m/G_m$
> Orthotropic: 9 constants; transversely isotropic: 5; isotropic: 2; anisotropic: 21.
> $[S] = [Q]^{-1}$; $[\bar{Q}] = [T]^{-1}[Q][R][T][R]^{-1}$; engineering constants E_x, E_y, G_{xy}, ν_{xy} vs θ.

Likely exam questions

> Derive E_1, E_2, ν_{12} and G_{12} using the rule of mixtures.
> Explain Hooke's law for different types of materials; number of independent elastic constants.
> Derive 2-D relationship of compliance and stiffness matrices for a lamina.
> Derive stress-strain relations for a lamina of arbitrary orientation; numerical problems.

Short Answer Questions (2 marks)

Q1. State the rule of mixtures for E_1.

Ans: $E_1 = E_fV_f + E_mV_m$ (equal-strain, upper bound).

Q2. Write the expression for the transverse modulus.

Ans: $1/E_2 = V_f/E_f + V_m/E_m$ (equal-stress, lower bound).

Q3. How many independent elastic constants are needed for an orthotropic lamina in plane stress?

Ans: Four: E_1, E_2, ν_{12}, G_{12}.

Q4. Define ν_{12}.

Ans: Ratio of transverse strain (direction 2) to longitudinal strain (direction 1), with sign change, when loaded in direction 1.

Q5. What is the physical meaning of $\bar{Q}_{16} \neq 0$?

Ans: Shear-extension coupling: normal strain produces shear stress in an off-axis lamina.

Q6. What is the Reuter matrix?

Ans: $[R] = \text{diag}(1, 1, 2)$ converting tensor shear strain to engineering shear strain.

Q7. Why is $[Q] \neq [S]^{-1}$ in 3-D but equal in plane stress?

Ans: In plane stress, the reduced stiffness $[Q]$ is the inverse of the 3×3 in-plane compliance $[S]$; the 3-D $[C]$ is a 6×6 matrix.

Q8. Number of constants in isotropic and anisotropic materials?

Ans: 2 and 21 respectively.

Q9. At which angle is the shear modulus G_{xy} of a lamina maximum?

Ans: Around 45°.

Practice Problems with Answers

Problem A

Carbon/epoxy $V_f = 0.6$: $E_f = 230$, $E_m = 3.5$ GPa, $\nu_f = 0.2$, $\nu_m = 0.35$. Find E_1, E_2, ν_{12}.

$E_1 = 230(0.6) + 3.5(0.4) =$ **139.4 GPa** ; $E_2 = 230 \times 3.5/(230 \times 0.4 + 3.5 \times 0.6) = 805/94.1 =$ **8.55 GPa** ; $\nu_{12} = 0.2(0.6) + 0.35(0.4) =$ **0.26**.

Problem B

Epoxy ($\rho = 1.2$) with E-glass ($\rho = 2.5$): fibre weight fraction 0.5. Find V_f.

$V_f = (0.5/2.5)/(0.5/2.5 + 0.5/1.2) = 0.2/(0.2 + 0.4167) =$ **0.324**.